Multi-Step Agent Repair: Which Restart Strategy Wins?

Task → Agent Execution → Step-wise Uncertainty Estimation → Predicted Failure Point → {Targeted Repair / Full Restart / Random Repair / Oracle Repair} → Evaluation & Causal Analysis

Motivation

Multi-step AI agents often fail because of a single mistake at one intermediate step. The common response is to throw away the whole trajectory and restart from scratch, which is expensive. A cheaper alternative is to fix only the broken step. But to fix a single step, you first need to find it — and a natural, cheap signal is the agent's own uncertainty. This work asks whether step-level repair actually beats full restart, and whether uncertainty is good enough to pick the step to fix.

Core Idea

We compare different ways of recovering from a failed agent trajectory and measure which is most effective and least expensive. We treat the step with the highest uncertainty as the *predicted* failure point, but we deliberately separate two questions:

(1) Repairing a single step is better than restarting everything?

(2) Can uncertainty correctly identify *which* step to repair?

Method

For each failed agent trajectory:

1. Estimate uncertainty at every intermediate step.
2. Mark the highest-uncertainty step as the predicted failure point.
3. Compare four repair strategies under a matched compute budget:
    - **Full Restart** — re-execute the entire trajectory.
    - **Random-Step Repair** — re-execute a randomly chosen step (lower-bound baseline).
    - **Uncertainty-Targeted Repair** — re-execute only the highest-uncertainty step (the practical method).
    - **Oracle-Targeted Repair** — re-execute the *truly* broken step using ground-truth labels (upper bound).
4. For every strategy, record both task success and computational cost (tokens/tool calls/steps).

Research Questions

- RQ1: Is targeted single-step repair more effective and efficient than restarting the whole trajectory? (Oracle vs. Full Restart)
- RQ2: How much performance is lost because uncertainty picks the wrong step? (Uncertainty-Targeted vs. Oracle)
- RQ3: Does uncertainty-based targeting beat random targeting at all? (Uncertainty-Targeted vs. Random)

Expected Outcome & Analysis

We expect oracle-targeted repair to beat full restart at lower cost, confirming that single-step repair is worthwhile *in principle*. The interesting result is the gap between oracle and uncertainty-targeted repair: this measures how reliable uncertainty really is as a localizer. The causal analysis then characterises *when* uncertainty points to the right step and *when* it fails (e.g., uncertain-but-correct exploratory steps, or errors that occur upstream of the uncertainty peak), giving a practical rule for when to trust targeted repair versus when to fall back to restart.